Project 0 — Bake-Off Report

Team: Nadia Ben Njima, Ahmed Brini, Rahma Sbei, Mariem Lahmar · **Date:** 2026-09-13

1. The task

We evaluate three models on 50 short math problems (arithmetic, algebra, word problems, basic finance), one model call per item, scored automatically by exact/symbolic equivalence (SymPy) against a human-verified answer.

Example item:

> Q: "A rectangle's length is 3m longer than width w. Perimeter is 26m. Calculate width w." Expected answer: `5`

2. The data

- **Source:** hand-written by the team, covering 3 difficulty bands (10 arithmetic, 10 algebraic substitution, 30 word problems / finance).
- **Label check:** Two team members independently solved each of the 50 items and verified that all expected ground-truth answers matched exactly.
- **Dev/test split:** No separate dev set; all 50 items served as the single frozen test set, and the system prompt was written and fixed before any model was executed.

3. Setup

Tier	Model	Provider
Top API	`openai/gpt-oss-120b`	Groq
Cheap API	`openai/gpt-oss-20b`	Groq
Local (self-hosted)	`qwen2.5:7b`	Ollama

Temperature 0, max_tokens 256, identical system prompt for all three models (forces `\boxed{answer}` -only output), same 50 items in the same order, same parser. Run on 13/9/2026.

4. Results

Tier	Model	Score	Accuracy	p50 latency	p95 latency	$/1k now	$/1k @100x
Top API	openai/gpt-oss-120b	50/50	100.0%	518.25 ms	2864.24 ms	$0.0693	$0.0693
Cheap API	openai/gpt-oss-20b	50/50	100.0%	565.54 ms	2493.64 ms	$0.0360	$0.0360
Local	qwen2.5:7b	24/50	48.0%	2094.75 ms	3233.62 ms	$0.00	$0.0000

3. wrong answers per model (from `results/per_item.csv`):

- Top: None — 0 errors (100.0% accuracy)
- Cheap: None — 0 errors (100.0% accuracy)

- Local: item `alg_04` — expected `12` , got `14` ; item `word_11` — expected `45` , got `\boxed{45\text{ meters}}` ; item `fin_02` — expected `150` , got `125`

5. Our choice

We recommend deploying **`openai/gpt-oss-20b`** (Cheap API) for production. It achieves perfect accuracy (100.0%), matching the top-tier 120b model, while cutting API costs nearly in half ($0.036 vs. $0.0693 per 1k requests) and offering lower p95 latency (2493.64 ms vs. 2864.24 ms). The local `qwen2.5:7b` model is unviable for this task due to its inadequate 48.0% accuracy and higher p50 latency (2094.75 ms). We would only switch to local self-hosting if traffic scales massively and fine-tuning bridges the accuracy gap, or upgrade to `openai/gpt-oss-120b` if problem complexity increases beyond the 20b model's reasoning baseline.

6. Cost at scale

- At 100x today's traffic: API costs remain flat at **$0.0693/1k** for Top API and **$0.0360/1k** for Cheap API due to pay-per-token tier pricing. The local model's effective cost drops to **$0.0000/1k** as fixed hardware amortization spreads across high request volume.
- **Break-even volume:** local hosting becomes as cheap as `openai/gpt-oss-20b` at ~72 requests/hour. Below that, the API is cheaper because you're paying for idle local hardware time.

Hardware note — local model (qwen2.5:7b via Ollama)

- **Machine:** [GPU: NVIDIA GeForce RTX 3050 , RAM: 24GB]
- **Ollama version:** [ollama version is 0.34.0]
- **Measured throughput:** p50 latency = [2094.75] ms, tokens/sec = [3.2] (from `results/summary.csv`)